A

Mini Project – II

Presentation

on

“Online Chatbot based Ticketing System”

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Presentation Outline

- Introduction
- Literature Survey
- Methodology(Hardware & Software)
- Schematic
- Flowchart /Algorithm
- Simulation Results (if any)
- Applications
- Conclusions
- References

Introduction

The project aims to design and develop an intelligent ticketing system using a chatbot interface. The system will be capable of understanding user queries, providing information, and automating ticketing processes

Introduction (Cont'd…..)

1. Need for Automation

   Traditional ticketing systems often involve manual processes, which can be time-consuming and prone to errors.

2. Enhanced User Experience

   A chatbot interface can provide a more user-friendly experience, allowing customers to access ticketing services easily.

3. Real-time Support

   The chatbot can provide instant support to users, addressing inquiries and resolving issues in real-time.

Literature Survey

S.No	Paper Title & Authors	Methodology	Key Contributions	Limitations	Year
1.	Keystroke- Level Model to Evaluate Chatbot Interface for Reservation System	Literature review of chatbot architectures for customer support. Comparison of frameworks like Dialog flow and Rasa.	Provides a high-level overview of chatbot architecture for automating ticket creation and management in customer support systems.	Lacks detailed implementation and real-world application data.	2019
2.	Leveraging AI Chatbots for IT Help Desk Automation: A Case Study	AI-driven chatbot implementation for automating help desk functions, including ticket generation and query resolution.	Demonstrates the use of NLP for classifying and routing support tickets, improving efficiency in ticket handling.	No focus on the chatbot interface or user interaction model. Limited to ticket classification.	2019
3.	Design and development of ticket booking system using smart bot	The main use of the chatbot on the website is to respond to the user's queries. For instance, the chatbot in messenger, and telegram bot. it is going to take text as input. once the	Provides a complete framework from chatbot design to deployment, with a focus on real-time ticket handling.	Primarily focused on architecture without extensive performance benchmarks.	2022

S.No	Paper Title & Authors	Methodology	Key Contributions	Limitations	Year
4.	Proposing Tourism Chatbot by employing the wisdom of crowds in building its knowledge base	The methodology used in this study was quantitative with the main activity was by conducting online questionnaires with closed-ended questions. The sampling method was non probabilistic with n=40 in the initial attempt	Detailed insights on how chatbots can improve efficiency in ticket handling and customer interaction processes.	Based on a single case study, so generalizability to other industries may be limited.	2022
5.	Enhanced Opened and distance learning using AI-Powered chatbot a conceptual framework	The proposed system will make use of bot Rule-based chatbots, AI Chatbots and Multinomial Naive Bayes algorithm. The rule-based bot completes scripted actions based on keywords and guided by a decision tree, a list of predefined options leading to the desired response are provided to the learner	Proposes AI chatbot solutions that reduce human intervention in help desk ticketing systems and automate ticket lifecycle processes.	Focuses on IT-specific help desk automation without generalization to other support systems.	2022
6.	Intelligent Agent based Ticket Booking System	This section discusses about the methodology used in developing the proposed system. The methodology chosen is	Highlights user experience challenges in using chatbots for IT ticketing and	Limited scope of chatbot functions—focused mainly on usability rather than full integration	2024

Methodology

Hardware :-
Windows 10/11, i3/i5 Core Processor, Internet

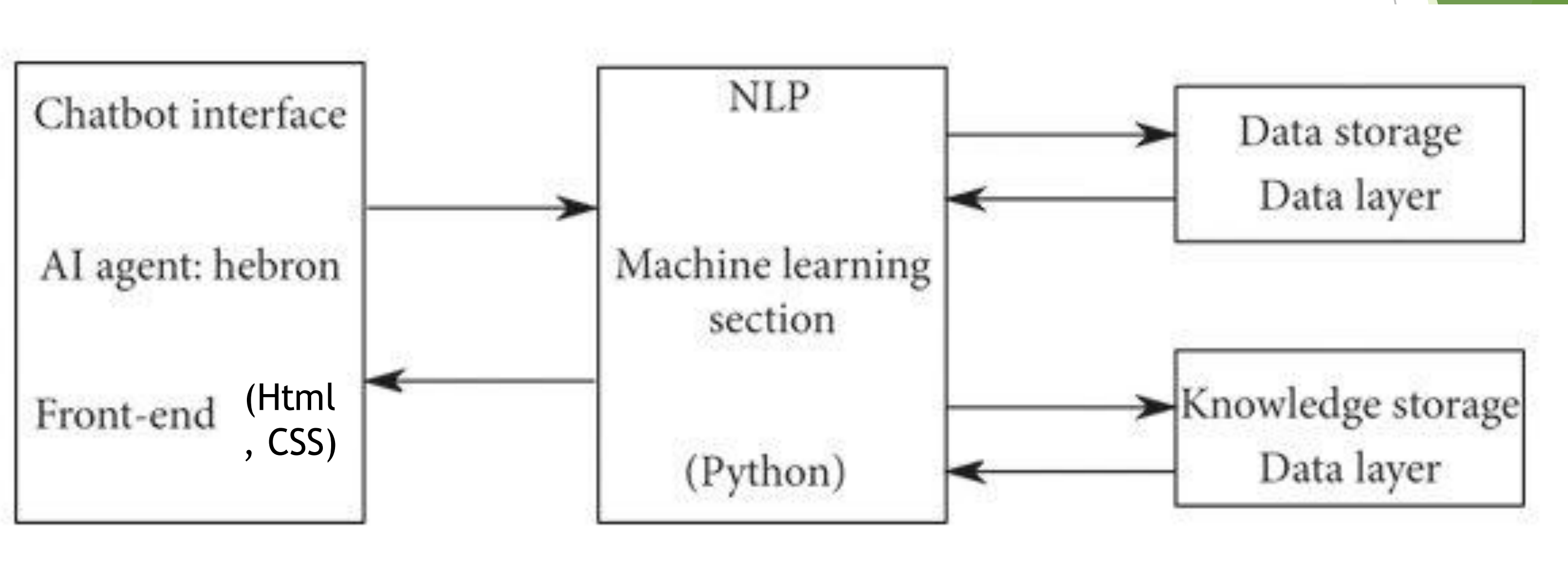
Software:-

VSCode, MYSQL, Browser.

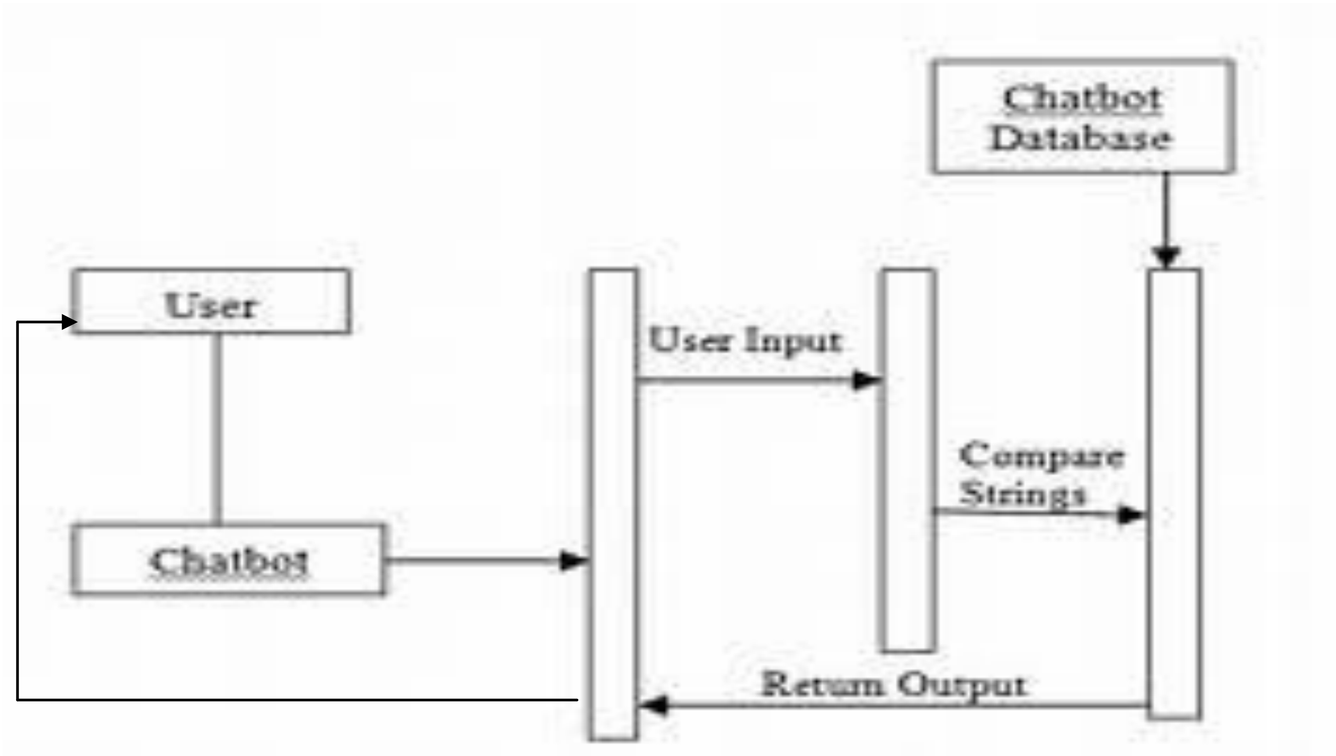
Technologies:-

HTML, CSS, Javascript, Python, Machine Learning, MYSQL.

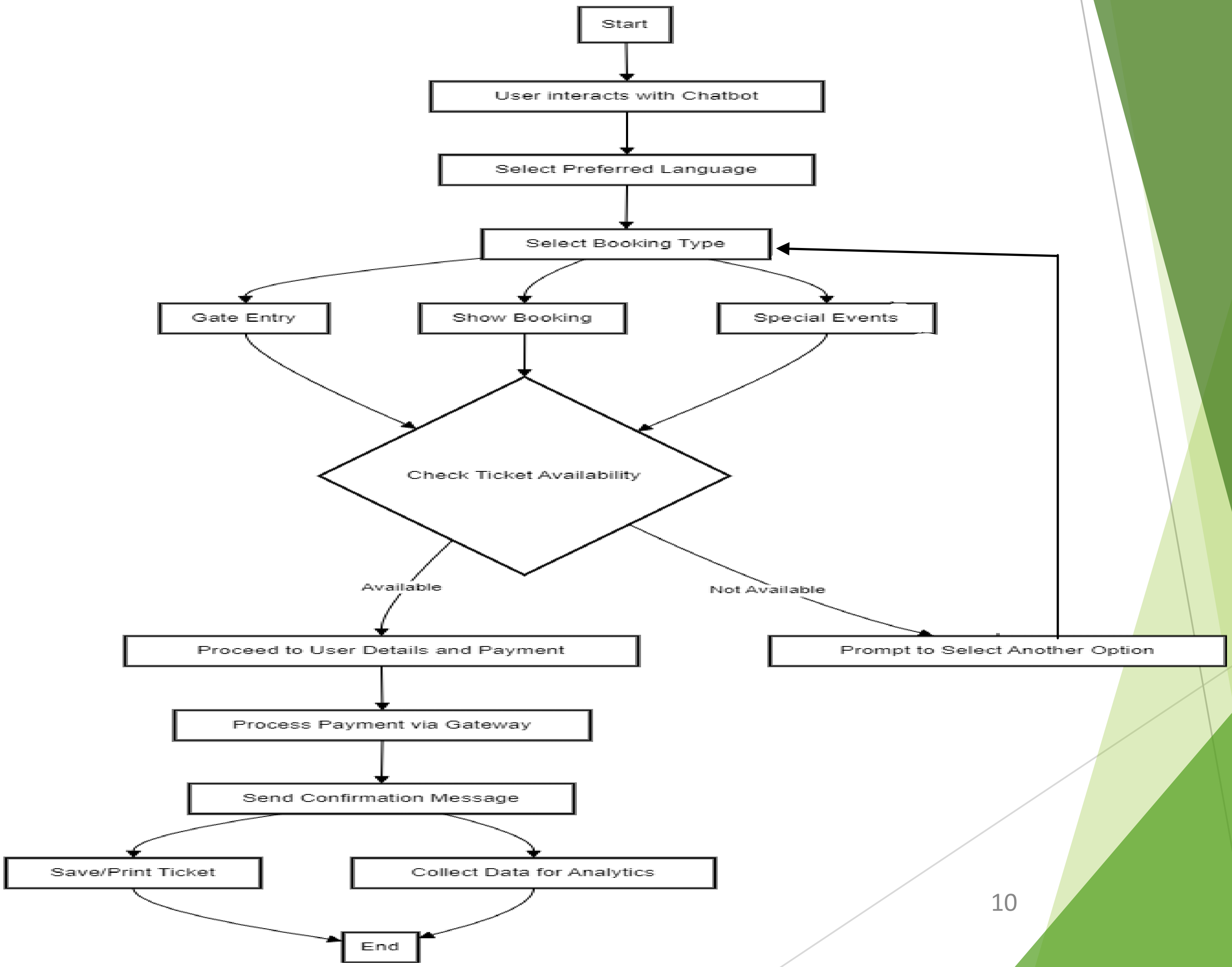
Block Diagram



Schematic -


Chatbot Database
User
User Input
Compare Strings
Chatbot
Return Output

Flowchart / Algorithm



Applications

1. **Event Ticketing**

   Streamline the purchase of tickets for concerts, sporting events, and other entertainment.

2. **Travel Booking**

   Seamlessly reserve flights, hotels, and rental cars through an intuitive chat interface.

3. **Public Transportation**

   Easily purchase and manage public transportation tickets, from subway to buses.

4. **Customer Service**

   Provide instant and personalized support to customers, resolving issues quickly.

Conclusion

1

Efficiency

Chatbot-based ticketing systems simplify ticket purchase and management, saving time and effort.

2

Personalization

Tailored recommendations based on user preferences and past purchases enhance the experience.

3

Accessibility

Easy-to-use chat interfaces are accessible to a wide range of users, regardless of technical skills.

4

Cost Savings

Automation reduces operational costs for ticketing organizations, improving profitability.

References

[1] Smirnov, A. Kashevnik, N. Shilov, N. Teslya and A. Shabaev(2014), Mobile application for guiding tourist activities: tourist assistant - TAIS, Proceedings of 16th Conference of Open Innovations Association FRUCT, Oulu, Finland, doi: 10.1109/FRUCT.2014.7000931

[2] U. Shanmugam, S. Mani, S. Sivakumar and P. Rajeswari, "Human-Computer text conversation through NLP in Tamil using Intent Recognition," 2019 International Conference on Vision Towards Emerging Trends in Communication and Networking (ViTECoN), Vellore, India, 2019, pp. 1-5, doi: 10.1109/ViTECoN.2019.8899571.

[3] J. D. Preece and J. M. Easton, Blockchain Technology as a Mechanism for Digital Railway Ticketing, IEEE International Conference on Big Data (Big Data), Los Angeles, CA, USA, 2019 doi: 10.1109/BigData47090.2019.9006293.

[4] E. Handoyo, M. Arfan, Y. A. A. Soetrisno, M. Somantri, A. Sofwan and E. W. Sinuraya, "Ticketing Chatbot Service using Serverless NLP Technology," 2018 5th International Conference on Information Technology, Computer, and Electrical Engineering (ICITACEE), Semarang, Indonesia, 2018, pp. 325-330, doi: 10.1109/ICITACEE.2018.8576921.

[5] Xiao Yu. (2021), Design of Aerospace QR Ticketing System Based on Mobile Devices,4th International Conference on Information Systems and Computer Aided Education (ICISCAE 2021). Association for Computing Machinery, 2021.https://dl.acm.org/doi/10.1145/3482632.3484146

[6] Abhay Tiwari and Ashima Mehta(2023), Flight Reservation System,International Journal of Advanced Research in Science, Communication and Technology (IJARSCT), ,DOI: 10.48175/IJARSCT-9152.

[7] Sai Chaithanya Gandhi.P,Sreenivasulu.G(2022), Flight Reservation System, International Journal of Creative Research Thoughts(IJCRT), Volume 10, Issue 6 June 2022 | ISSN: 2320-2882.

[8] Majduddeen Al Jalaliya P.P, Dr. Gajraj Singh Ahirwar(2018), A Study of E-Ticketing System and Its Value in Indian Railways, International Journal of Advance Research and Innovative Ideas in Education, IJARIIE-ISSN(O)-2395-4396

Thank you!